

Emir Yılmaz

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Research Interests	Tribology, Surface Engineering, Heat Transfer, Machine Learning	
Current Appointment	Sophia University Associate Professor Department of Engineering & Applied Sciences <i>Precision Engineering Research Group</i>	Tokyo, Japan 2026 - present
Education	Sophia University Ph.D., Mechanical Engineering Dissertation: "Development of Heat Transfer Models for Intake Systems" Advisors: Takashi Suzuki, Mitsuhsa Ichiyanagi M.Sc., Mechanical Engineering	Tokyo, Japan 2020 2017
	Sabanci University B.Sc., Mechatronics Engineering, <i>minor in Energy Systems</i> Advisor: Serhat Yeşilyurt Study abroad semester at <i>Tokyo Metropolitan University</i>	Istanbul, Türkiye 2014 Spring 2013
Academic Experience	Sophia University , Dept. of Engineering & Applied Sciences Assistant Professor Postdoc Fellow, <i>Thermal Engineering Lab.</i> Research Assistant Teaching Assistant	Tokyo, Japan 2021 - 2026 2020 - 2021 2018 - 2020 2016 - 2018
Industry Experience	Volvo Group Trucks Technology , Ageo R&D Plant Vehicle Dynamics Engineering Intern Ford Motor Company , Yeniköy Assembly Plant Project & Maintenance Engineer, <i>Six-Sigma Green Belt</i>	Saitama, Japan Feb-March 2016 Kocaeli, Türkiye 2014-2015
Publications	Up-to-date peer-reviewed published papers are available here .	
Grants	ISUZU Advanced Engineering Center, LTD. Surface micro-texturing effect on friction loss reduction in reciprocating sliding contacts, Budget: ¥2,200,000 (Co-PI) Sophia University - FY2025 FST Competitive Research Grant Enhancement of Pool Boiling Heat Transfer through Gas-Liquid Behavior Control Using Functional Surfaces, Budget: ¥4,110,000 (PI)	2021 - 2025 2025

Honors, Awards & Scholarships	Graduate School Research Award (JSAE)	2019
	Best Paper & Best Presentation Awards, International Conference of Automotive Technology for Young Engineers (ICATYE 2017)	2018
	MEXT Scholarship, Ph.D. (Japanese Government)	2017
	Nikki-Sayenoshi Scholarship, M.Sc. (Private Company)	2016
	JASSO Honors Scholarship, M.Sc. (Japanese Government)	2015
	Youth Exchange Ambassadorial Scholarship (Rotary International)	2008
Teaching	Sophia University , Dept. of Engineering & Applied Sciences <i>Tokyo, Japan</i>	
	MAT 102: Linear Algebra	Spring'26
	MAT 104: Basics of Differential Equations	Fall'21 - Present
	MEC 211: Basics of Mechanical System Design	Spring'21 - Spring'23
	MEC 213: Applied Mechanics	Fall'25
	MEC 293: Mechanical System Exercise I	Spring'24 - Present
	MEC 212: Special Topics on Mechanical Engineering	Fall'21 - Present
	MEC 361: Machine Design	Fall'21 - Present
	MEC 514: Fundamentals of Microsystem Design	Spring'21 - Spring'23
	MEC 514: Fundamentals of Tribology	Spring'24 - Present
Invited Talks	ADS 561: Machine Design & Data Analysis	Spring'24 - Present
	The future is here: low-carbon society & hydrogen-based fuels	2021
	Two mock lectures for high school students during the Open Campus event in Sophia University (in Japanese).	
	Development of Carbon Free Ammonia Engine	2021
Skills	Presented on the development of ammonia internal combustion engines at the Zero Emission Mobility Power Source Research Consortium (in Japanese).	
	Towards Zero Emission Transportation: The Role of Ammonia in Carbon-Free Internal Combustion Engine Development	2024
	The 2nd IITD-Sophia Workshop on Transportation Science & Technology	
Professional Memberships	Software:	
	GT-Power, Ansys Fluent, CAD (Creo, SolidWorks, Fusion 360)	
	Programming:	
	FORTRAN, Python, MATLAB; some experience with C++, JavaScript, $\LaTeX$	
	Languages:	
	English (advanced), Japanese (fluent), Turkish (native)	
	Society of Automotive Engineers of Japan (JSAE)	2016 – Present
	The Japan Society of Mechanical Engineers (JSME)	2021 – Present
	The Japan Society for Precision Engineering (JSPE)	2021 – Present
	The Japanese Society of Tribologists (JAST)	2022 – Present